Data Exploration

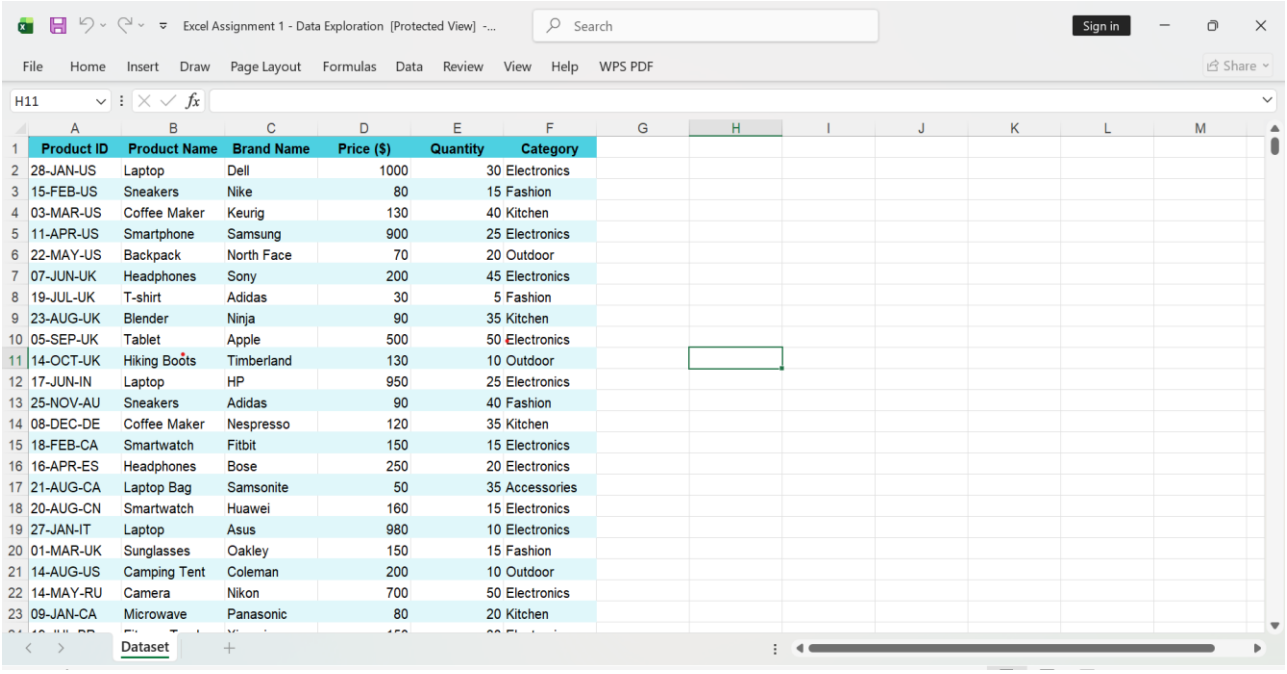
Excel Assignment-1

DATA EXPLORATION:

Definition:

Data exploration is the initial process of analyzing a dataset to understand is structure, values, and patterns.It helps in identifying key information and gaining insights from the data.

In this assignment, the product dataset is used to perform basic statistical Operations.



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen

Observation:

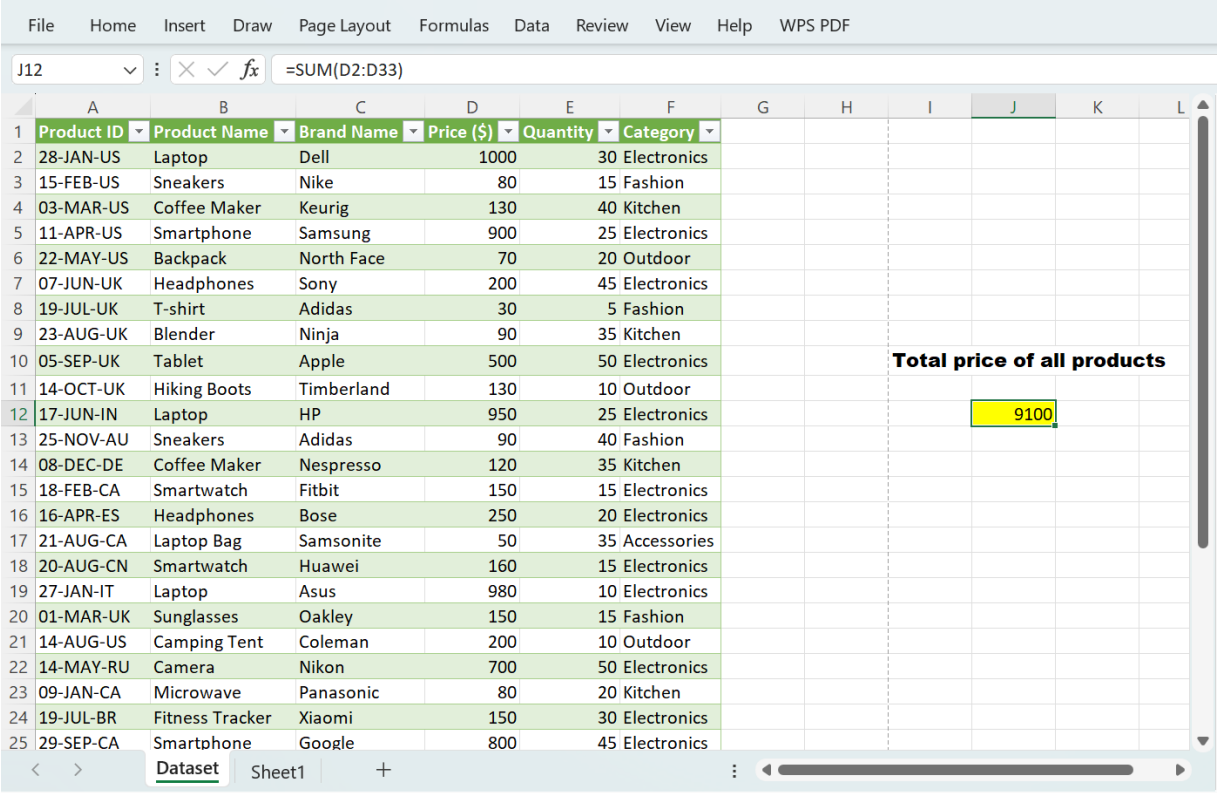
1. The dataset contains 35 rows and 6 columns, representing structured product data.

2. Data cleaning is the process of preparing the dataset by removing errors, inconsistencies, and unwanted data.

3. In the given dataset, there are 32 rows and 6 columns. During data cleaning, 2 duplicate rows were identified. (Example Laptop bag, Laptop)

4. The dataset is well-structured with no missing values visible, making it suitable for analysis.

➢ SUM FUNCTION =SUM(D2:D33)



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
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19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
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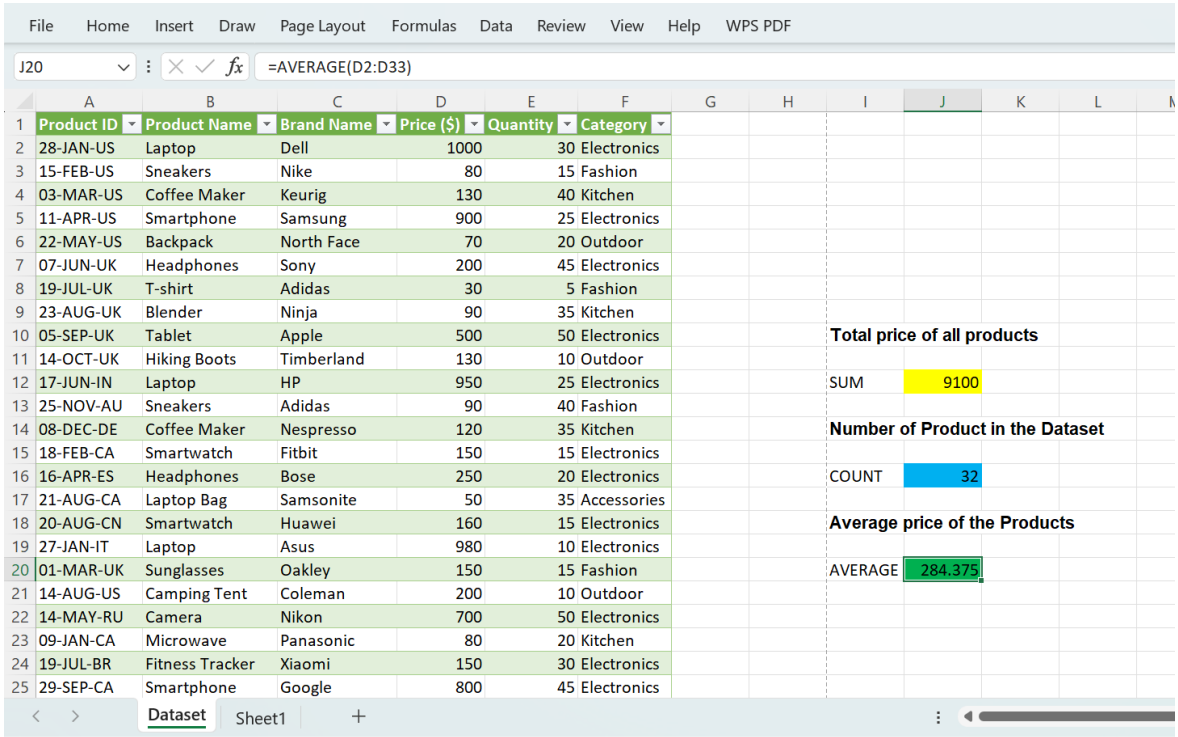
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
29-SEP-CA	Smartphone	Google	800	45	Electronics

➢ COUNT FUNCTION =COUNTA(D2:D33)



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19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
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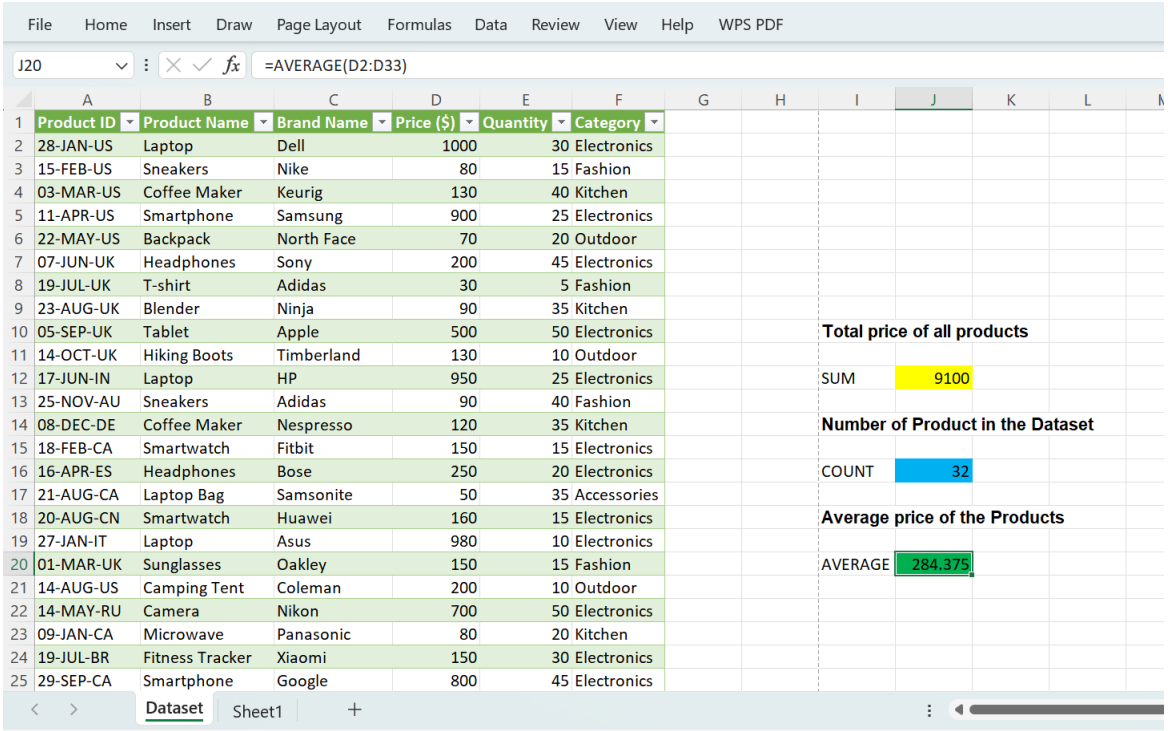
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
29-SEP-CA	Smartphone	Google	800	45	Electronics

➢ AVERAGE PRICE =AVERAGE(D2:D33)



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
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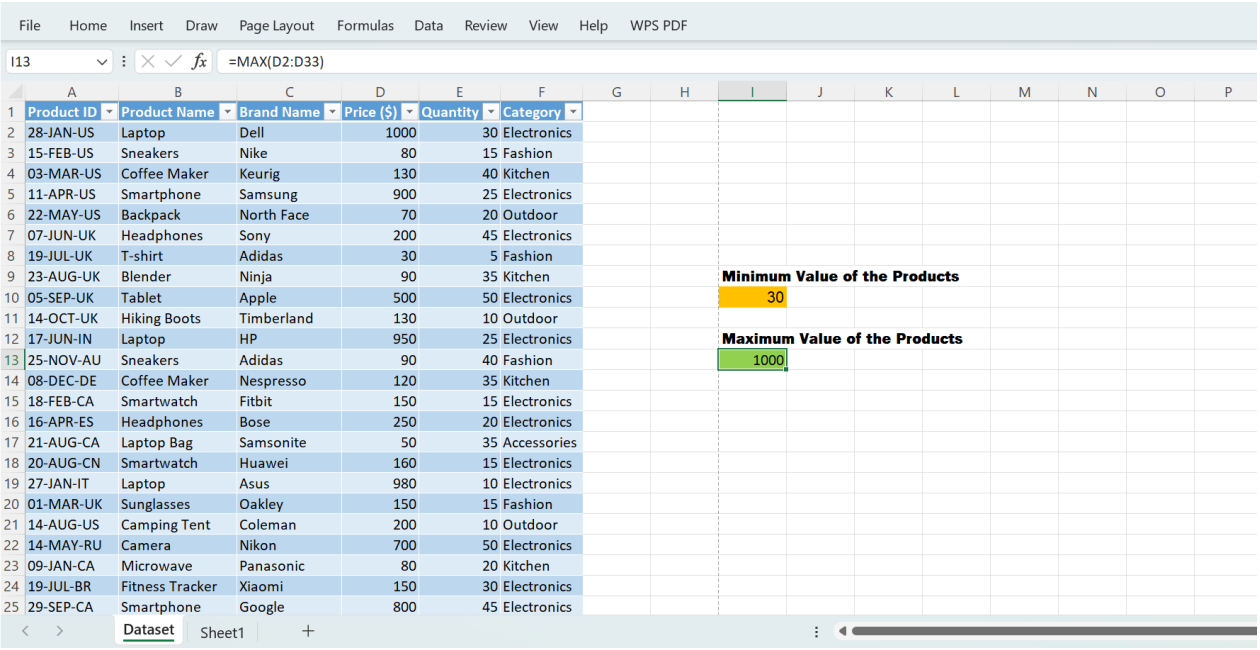
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
29-SEP-CA	Smartphone	Google	800	45	Electronics

<u>MINIMUM AND MAXIMUM VALUES</u>

- ❖ MIN FUNCTION =MIN(D2:D33)
- ❖ MAX FUNCTION =MAX(D2:D33)



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03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
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19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
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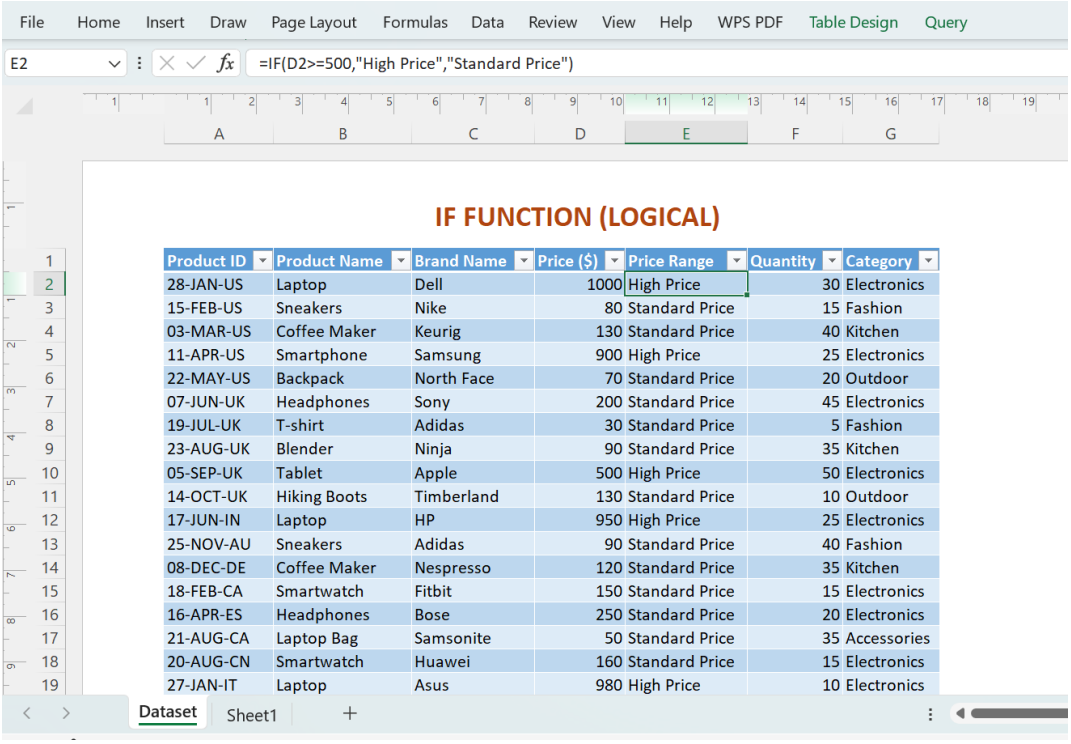
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
29-SEP-CA	Smartphone	Google	800	45	Electronics

LOGICAL FUNCTION -IF

IF FUNCTION =IF(D2>=500,”High price”,”Standard price”)



Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
28-JAN-US	Laptop	Dell	1000	High Price	30	Electronics
15-FEB-US	Sneakers	Nike	80	Standard Price	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	Standard Price	40	Kitchen
11-APR-US	Smartphone	Samsung	900	High Price	25	Electronics
22-MAY-US	Backpack	North Face	70	Standard Price	20	Outdoor
07-JUN-UK	Headphones	Sony	200	Standard Price	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	Standard Price	5	Fashion
23-AUG-UK	Blender	Ninja	90	Standard Price	35	Kitchen
05-SEP-UK	Tablet	Apple	500	High Price	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	Standard Price	10	Outdoor
17-JUN-IN	Laptop	HP	950	High Price	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	Standard Price	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	Standard Price	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	Standard Price	15	Electronics
16-APR-ES	Headphones	Bose	250	Standard Price	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard Price	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	Standard Price	15	Electronics
27-JAN-IT	Laptop	Asus	980	High Price	10	Electronics

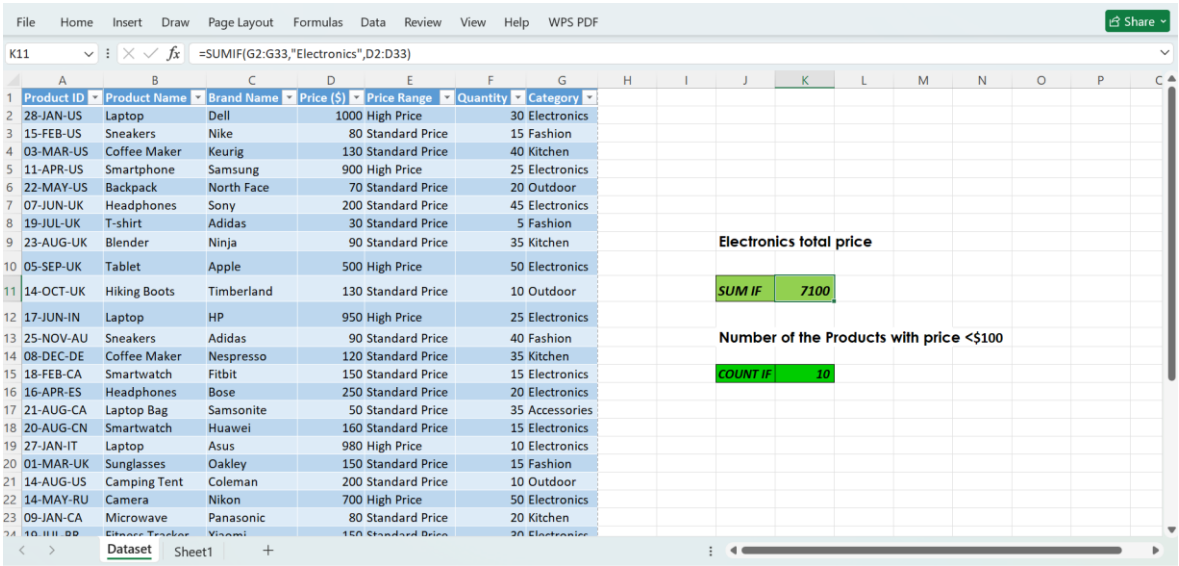
Explanation:

- If the price (D2) is greater than or equal to 500⟶result will be “high price”.
- If the price is less than 500⟶result will be “standard price”.

Conditional Functions - SUMIF & COUNTIF

1. SUMIF FUNCTION =(G2:G33,”Electronics”,D2:D33)

2. COUNTIF FUNCTION =(D2:D33,”<100”)



TEXT FORMATTING - LEFT, RIGHT, MID

- LEFT FUNCTION = LEFT(D2:D33,2)

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A2 =LEFT(D2:D33,2)

	A	B	C	D	E	F	G	H	I	J
1	Day	Month	Country Code	Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
2	28	JAN	US	28-JAN-US	Laptop	Dell	1000	High Price	30	Electronics
3	15	FEB	US	15-FEB-US	Sneakers	Nike	80	Standard Price	15	Fashion
4	03	MAR	US	03-MAR-US	Coffee Maker	Keurig	130	Standard Price	40	Kitchen
5	11	APR	US	11-APR-US	Smartphone	Samsung	900	High Price	25	Electronics
6	22	MAY	US	22-MAY-US	Backpack	North Face	70	Standard Price	20	Outdoor
7	07	JUN	UK	07-JUN-UK	Headphones	Sony	200	Standard Price	45	Electronics
8	19	JUL	UK	19-JUL-UK	T-shirt	Adidas	30	Standard Price	5	Fashion
9	23	AUG	UK	23-AUG-UK	Blender	Ninja	90	Standard Price	35	Kitchen
10	05	SEP	UK	05-SEP-UK	Tablet	Apple	500	High Price	50	Electronics
11	14	OCT	UK	14-OCT-UK	Hiking Boots	Timberland	130	Standard Price	10	Outdoor
12	17	JUN	IN	17-JUN-IN	Laptop	HP	950	High Price	25	Electronics
13	25	NOV	AU	25-NOV-AU	Sneakers	Adidas	90	Standard Price	40	Fashion
14	08	DEC	DE	08-DEC-DE	Coffee Maker	Nespresso	120	Standard Price	35	Kitchen
15	18	FEB	CA	18-FEB-CA	Smartwatch	Fitbit	150	Standard Price	15	Electronics
16	16	APR	ES	16-APR-ES	Headphones	Bose	250	Standard Price	20	Electronics
17	21	AUG	CA	21-AUG-CA	Laptop Bag	Samsonite	50	Standard Price	35	Accessories
18	20	AUG	CN	20-AUG-CN	Smartwatch	Huawei	160	Standard Price	15	Electronics
19	27	JAN	IT	27-JAN-IT	Laptop	Asus	980	High Price	10	Electronics
20	01	MAR	UK	01-MAR-UK	Sunglasses	Oakley	150	Standard Price	15	Fashion
21	14	AUG	US	14-AUG-US	Camping Tent	Coleman	200	Standard Price	10	Outdoor
22	14	MAY	RU	14-MAY-RU	Camera	Nikon	700	High Price	50	Electronics
23	09	JAN	CA	09-JAN-CA	Microwave	Panasonic	80	Standard Price	20	Kitchen
24	19	JUL	BR	19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard Price	30	Electronics
25	29	SEP	CA	29-SEP-CA	Smartphone	Google	800	High Price	45	Electronics

Dataset

- MID FUNCTION =MID(D2:D33,4,3)

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B2 =MID(D2:D33,4,3)

	A	B	C	D	E	F	G	H	I	J	K
1	Day	Month	Country Code	Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category	
2	28	JAN	US	28-JAN-US	Laptop	Dell	1000	High Price	30	Electronics	
3	15	FEB	US	15-FEB-US	Sneakers	Nike	80	Standard Price	15	Fashion	
4	03	MAR	US	03-MAR-US	Coffee Maker	Keurig	130	Standard Price	40	Kitchen	
5	11	APR	US	11-APR-US	Smartphone	Samsung	900	High Price	25	Electronics	
6	22	MAY	US	22-MAY-US	Backpack	North Face	70	Standard Price	20	Outdoor	
7	07	JUN	UK	07-JUN-UK	Headphones	Sony	200	Standard Price	45	Electronics	
8	19	JUL	UK	19-JUL-UK	T-shirt	Adidas	30	Standard Price	5	Fashion	
9	23	AUG	UK	23-AUG-UK	Blender	Ninja	90	Standard Price	35	Kitchen	
10	05	SEP	UK	05-SEP-UK	Tablet	Apple	500	High Price	50	Electronics	
11	14	OCT	UK	14-OCT-UK	Hiking Boots	Timberland	130	Standard Price	10	Outdoor	
12	17	JUN	IN	17-JUN-IN	Laptop	HP	950	High Price	25	Electronics	
13	25	NOV	AU	25-NOV-AU	Sneakers	Adidas	90	Standard Price	40	Fashion	
14	08	DEC	DE	08-DEC-DE	Coffee Maker	Nespresso	120	Standard Price	35	Kitchen	
15	18	FEB	CA	18-FEB-CA	Smartwatch	Fitbit	150	Standard Price	15	Electronics	
16	16	APR	ES	16-APR-ES	Headphones	Bose	250	Standard Price	20	Electronics	
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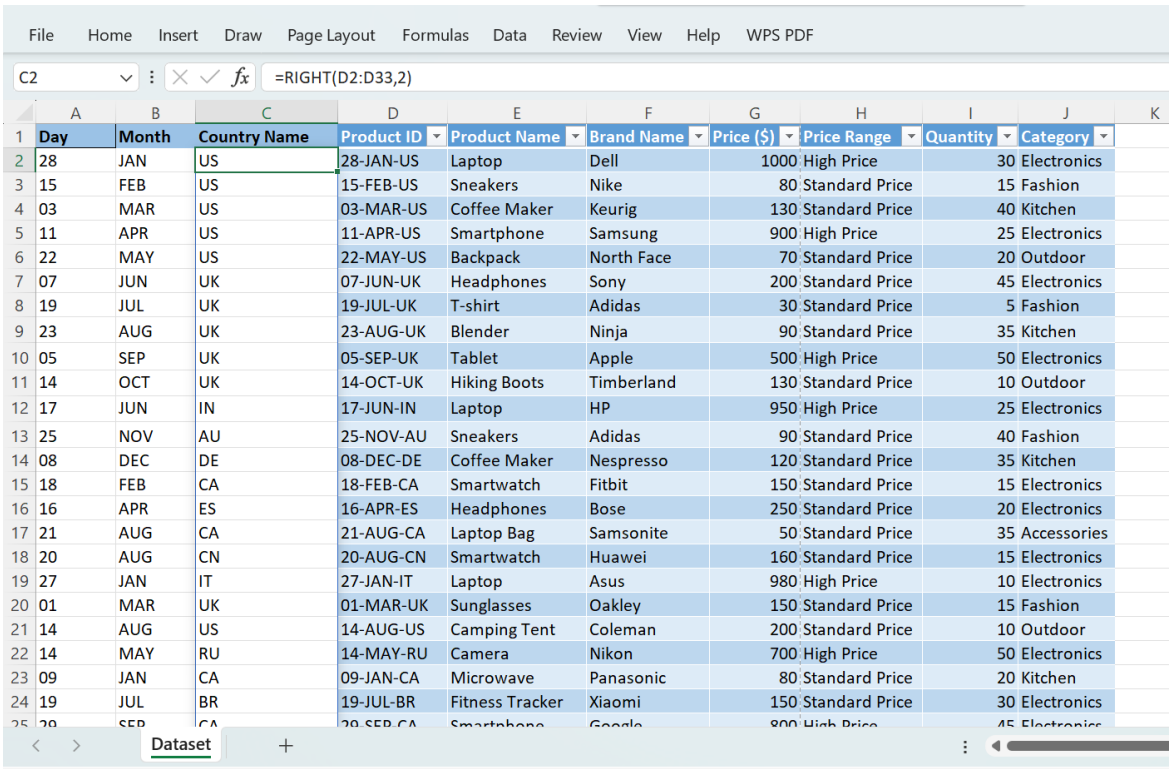
18	20	AUG	CN	20-AUG-CN	Smartwatch	Huawei	160	Standard Price	15	Electronics	
19	27	JAN	IT	27-JAN-IT	Laptop	Asus	980	High Price	10	Electronics	
20	01	MAR	UK	01-MAR-UK	Sunglasses	Oakley	150	Standard Price	15	Fashion	
21	14	AUG	US	14-AUG-US	Camping Tent	Coleman	200	Standard Price	10	Outdoor	
22	14	MAY	RU	14-MAY-RU	Camera	Nikon	700	High Price	50	Electronics	
23	09	JAN	CA	09-JAN-CA	Microwave	Panasonic	80	Standard Price	20	Kitchen	
24	19	JUL	BR	19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard Price	30	Electronics	
25	29	SEP	CA	29-SEP-CA	Smartphone	Google	800	High Price	45	Electronics	

Dataset

- RIGHT FUNCTION = RIGHT(D2:D33,2)



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C2 =RIGHT(D2:D33,2)

Day	Month	Country Name	Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
28	JAN	US	28-JAN-US	Laptop	Dell	1000	High Price	30	Electronics
15	FEB	US	15-FEB-US	Sneakers	Nike	80	Standard Price	15	Fashion
03	MAR	US	03-MAR-US	Coffee Maker	Keurig	130	Standard Price	40	Kitchen
11	APR	US	11-APR-US	Smartphone	Samsung	900	High Price	25	Electronics
22	MAY	US	22-MAY-US	Backpack	North Face	70	Standard Price	20	Outdoor
07	JUN	UK	07-JUN-UK	Headphones	Sony	200	Standard Price	45	Electronics
19	JUL	UK	19-JUL-UK	T-shirt	Adidas	30	Standard Price	5	Fashion
23	AUG	UK	23-AUG-UK	Blender	Ninja	90	Standard Price	35	Kitchen
05	SEP	UK	05-SEP-UK	Tablet	Apple	500	High Price	50	Electronics
14	OCT	UK	14-OCT-UK	Hiking Boots	Timberland	130	Standard Price	10	Outdoor
17	JUN	IN	17-JUN-IN	Laptop	HP	950	High Price	25	Electronics
25	NOV	AU	25-NOV-AU	Sneakers	Adidas	90	Standard Price	40	Fashion
08	DEC	DE	08-DEC-DE	Coffee Maker	Nespresso	120	Standard Price	35	Kitchen
18	FEB	CA	18-FEB-CA	Smartwatch	Fitbit	150	Standard Price	15	Electronics
16	APR	ES	16-APR-ES	Headphones	Bose	250	Standard Price	20	Electronics
21	AUG	CA	21-AUG-CA	Laptop Bag	Samsonite	50	Standard Price	35	Accessories
20	AUG	CN	20-AUG-CN	Smartwatch	Huawei	160	Standard Price	15	Electronics
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14	AUG	US	14-AUG-US	Camping Tent	Coleman	200	Standard Price	10	Outdoor
14	MAY	RU	14-MAY-RU	Camera	Nikon	700	High Price	50	Electronics
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19	JUL	BR	19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard Price	30	Electronics
29	SEP	CA	29-SEP-CA	Smartphone	Google	800	High Price	45	Electronics

Dataset

Definition:

- Using Excel functions like LEFT, RIGHT, and MID meaningful information was extracted from the product ID column.
- New columns such as Day, Month, and Country code were created, Making the dataset more useful.

CONCULSION:

Through data Cleaning and transformation, a better foundation for accurate analysis was achieved.